# Tomas Pereira - Machine Learning Engineer

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## PROFILE

As a Computer Science graduate specializing in AI, I am passionate and skilled in the research and development of machine learning models and applications. Inquisitive and thorough, I thrive on understanding systems at a fundamental level with a recent interest in Conversational AI and AI agents that perform tasks independently. I strive to merge my technical skills with my forward-thinking attitude to push the boundaries on what AI can achieve and create innovative tools that solve real-world problems.

## WORK EXPERIENCE

**Atlas Analytics Lab** **May. 2023– Dec 2024**
*Master's/NSERC Researcher* *Concordia University, Montreal, QC*

Early Member of the Research Lab, which is Focused on the Development of Deep Learning applications for Computational Pathology. Worked on the Research and Development of a Multimodal Vision-Language Model for Automatic Report Generation using the CLIP Framework and of a Foundational Model for Breast and Colon Cancer Histopathology Tissue.

- Recipient of the NSERC-USRA Research Award in May 2023
- Developed a Framework for Individual Histopathology Object Extraction using Meta's SAM Model
- Interdisciplinary Collaboration with Domain Experts in the Field of Pathology from the CHUM Hospital

**Concordia University** **Sept. 2023 - Present**
*Teaching Assistant* *Montreal, QC*

Teaching Assistant for the Machine Learning Course for both Graduate and Undergraduate Students across Multiple Semesters. Prepared Material and Held Weekly Labs to Teach the Implementation of Machine Learning.

- Developed Teaching Expertise in ML Theory, Pytorch, Numpy and Scikit-Learn
- Guided and Mentored Students in the Development of Their Own Deep Learning Projects
- Tasked with Grading for Theoretical and Practical Assignments, along with Examinations

**Bureau Veritas** **July. 2022 – May. 2023**
*Systems Specialist* *Montreal, QC*

Specialist of Digital Operation of Tasks in the Laboratory and Primary Contact for Data Related Tasks and Issues.

- Successful Refactoring of Datasheets and VBA Macros used in Daily Operations of Lab Analysts
- Performed Data Wrangling and Cleaning Tasks to Half the Time Required by QA Analysts to Vet Records
- Key Figure in Development and Adoption of a New Inventory System for Lab Supplies and Bio Samples

## EDUCATION

**Concordia University** **Sept. 2023 - May 2025**
*Master of Computer Science (MApCompSc)* *Montreal, QC*

- Specialization in AI, LLMs, Reactive Programming, and Client-Server Applications
- Recipient of the Concordia Merit Scholarship Award (2024 and 2025)
- Studied Representation Learning and Reinforcement Learning at MILA - Quebec AI Institute

**Concordia University** **Sept 2019 - May 2023**
*Bachelor of Computer Science (BCompSc)* *Montreal, QC*

- Recipient of the Government of Quebec Award under the Merit Scholarship Program (2021)

## PROJECTS

### MultiSpeech - A Transformer based Multi Speaker Text-to-Speech Model

Model and paper recreation of this 7M parameter TTS Transformer model. Developed and trained this system from scratch using the VCTK dataset with 109 different speakers.

- This model converts text into speech signals by generating Mel Spectrograms and decoding them with the HiFi-GAN vocoder.
- Integration of complex attention mechanisms to enable the model to learn high quality multi-modal alignment.

### Research Project - Updating Method-Level Comments Using Generative AI

This Project Tackled the Application of Generative AI for the Domain of Software Engineering. Two Models were developed with the Task of Updating Existing Code-Comments when the Relevant Code is Changed.

- Led the Effort to Fine Tune the CodeT5 Transformer and Google's Gemma (LLM) Model
- Studied the Effectiveness of Varied Inputs and Gradual Information Gain with Common Textual Similarity Benchmarks like BLEU and Meteor Score

### Research Project - Reinforcement Learning with Multi Instance Learning For Efficient WSI Analysis

Using Reinforcement Learning, this Project developed an Agent which could Detect the Tissue Segments on a Whole Slide Image more Efficiently than other Existing Pre-processing Methods

- Early work in the Application of RL for Computational Pathology, which is Mostly Unexplored for the Field
- Future Extension of this work is Aimed at Developing a Cancer Classifier Which Can Perform as Well as the State-of-the-art While Utilization Significantly Less Compute Power

### PLAY Framework Reactive System

This Project Developed a Reactive Application Interfacing the YoutubeV3 API using Asynchronous Message Passing. The Aim was to Learn the Concepts of Reactive Programming and Modern Java Development

- Led Group and Assigned Tasks to Team Members To Ensure Steady And Efficient Development
- Developed a Component for Sentiment Analysis Based on Video Descriptions

## SKILLS & INTERESTS

- **Technical Skills:**
  - Languages: Python, Java, SQL, VBA, C++, Ruby, C#
  - Technologies: Pytorch, Numpy, SKLearn, OpenAI API, HuggingFace, PLAY, SpeechBrain, OpenCV, Gymnasium, Pandas, SLURM
  - Other: Deep Learning, Neural Networks, R&D, Computer Vision, NLP, LLMs, Decision-Making Systems, Medical Imaging, Data Science, Backend Programming
- **Soft Skills:**
  - Problem Solving; Critical Thinking; Collaboration; Leadership; Communication; Analysis; Mentoring; Presentation; Innovation; Project Management; Attention to Detail; Adaptability; Creativity;
  - Spoken Languages: English, French and Portuguese (Trilingual)
- **Interests**: Conversational AI; Dungeons and Dragons; Game Development; Gardening; Rugby;

**REFERENCES**: Available Upon Request